Interview STAR Defense Guide

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Designed for natural, spoken-word interview delivery (conversational, structured, metric-backed).

Story 1: EMR Platform Lifecycle, Reliability & Runtime Upgrades

> *"Interview Soundbite: I didn't just bump version numbers; I dug into root-cause Iceberg and YARN plugin incompatibilities under the new runtime and turned ad-hoc patching into an automated, zero-downtime release cycle."*

Situation: We had multiple business-critical EMR clusters running on legacy runtimes that were approaching strict AWS end-of-support deadlines. If we missed the cutoff, we risked compliance penalties, security vulnerabilities, and loss of vendor support.

Task: My responsibility was to lead the end-of-support remediation across DEV, PRE, and PROD, ensuring all active pipelines transitioned seamlessly without breaking production data workflows.

Action (What I Did): Instead of treating it as a simple config change, I staged upgrades systematically across environments. When our target runtime broke the Lake Formation Iceberg fine-grained access control (FGAC) plugin and triggered YARN initialization exceptions, I personally investigated the dependency chain, updated the underlying IAM service roles, and reconfigured the plugin bindings. To prevent silent failures, I wired AWS Health Dashboard events directly into Amazon SNS alerts and authored standard operating runbooks so team on-call engineers could validate clusters instantly.

Result: We remediated 100% of end-of-support clusters well within our SLA windows with zero unplanned pipeline outages, cutting cluster incident recovery times significantly.

Story 2: Lake Formation Self-Service Access & Tag-Based Access Control (TBAC)

> *"Interview Soundbite: I eliminated human bottlenecks by deploying Lake Formation tag-based governance, turning days of waiting on access tickets into minutes of automated provisioning."*

Situation: As data consumers and analytics teams expanded, our data platform team was overwhelmed with manual tickets requesting access to S3 tables and databases. Requests took days to fulfill, slowing down analytics while increasing the risk of permission creep.

Task: I needed to decentralize and automate access control while strictly enforcing least-privilege security and multi-account compliance.

Action (What I Did): I architected and rolled out AWS Lake Formation Tag-Based Access Control (LF-Tags/TBAC) across our multi-account setup. Instead of managing complex per-table grants, I tagged data assets based on confidentiality and domain. I then engineered a self-service, web-based Lake Formation Access Automation workflow paired with standardized cross-account IAM role mapping guidelines, allowing pre-approved roles to inherit exact permissions dynamically.

Result: Access provisioning lead time dropped from several working days to under one hour, removing our team from the critical path and maintaining a 100% audit-compliant permission model.

Story 3: Scalable Team Onboarding & CI/CD Pipeline Modernization

> *"Interview Soundbite: I treated platform onboarding as a product—codified repository creation with Terraform, hardened pipeline security with Trivy, and cut onboarding lead time by 70%."*

Situation: Onboarding new producer and consumer teams required custom, manual configuration for every repository, IAM policy, and deployment role. At the same time, existing pipelines were failing intermittently due to brittle cross-account authentication and legacy vulnerability scanners.

Task: I owned the initiative to automate the provisioning lifecycle for dataset repositories and modernize our CI/CD security posture.

Action (What I Did): I built modular Terraform templates to provision dataset repositories, group permissions, and S3 folders automatically upon ticket trigger. In the CI/CD pipeline, I debugged and resolved intermittent AWS OIDC AssumeRole authentication issues across environments and led the security scanner migration from deprecated `tfsec` to `trivy`, establishing standardized linting and security gates.

Result: Team onboarding cycle time dropped by ~70%, while CI/CD vulnerability scanning reached 100% compliance across all active dataset repositories.

Story 4: Snowflake Governance & AWS Athena FinOps

> *"Interview Soundbite: I bridged platform performance and FinOps by putting hard spend guardrails on Athena queries and automating Snowflake historical metadata tagging."*

Situation: We saw rising ad-hoc query costs in AWS Athena and inconsistent data categorization in Snowflake, making it difficult for compliance to trace data lineage and track ownership across business lines.

Task: My objective was to enforce query cost controls on AWS Athena and build an automated metadata tagging mechanism in Snowflake.

Action (What I Did): For Athena, I configured dedicated workgroups with per-query and daily data scan quotas, while optimizing table partition patterns to curb runaway scans. For Snowflake, I configured Snowflake Data Exchange infrastructure and implemented automated, scheduled

historical tagging procedures across tables and views, embedding operational ownership and data classification directly into the data catalog.

Result: We prevented unbudgeted cloud spend spikes across analytics workgroups and achieved 100% data catalog compliance for enterprise audits.

Story 5: Common Data Framework (CDF) Architecture & Migration

> *"Interview Soundbite: I mapped out our Common Data Framework blueprints across batch and streaming, audited legacy dependencies, and gave data science teams the exact playbook to migrate safely."*

Situation: Multiple analytics and modeling teams across Risk, Compliance, and COO were running on legacy, siloed infrastructure with no unified path to transition onto the cloud data platform.

Task: I was tasked with establishing the enterprise integration architecture blueprints and creating a clear migration roadmap for analytics use cases.

Action (What I Did): I designed end-to-end architecture specifications for the Common Data Framework (CDF), establishing proven design patterns for batch ingestion, event streaming, and outbound SFTP delivery. I conducted comprehensive application and model inventory audits across business units to map dependencies, and authored the Data Science Migration Toolkit ("How-To" guide) to provide step-by-step instructions for refactoring workloads into the cloud.

Result: Standardized architectural blueprints across multiple business divisions and de-risked migration timelines, enabling legacy system decommission with zero operational disruption.